

Day 2: TCP/IP Suite - Assignment Answers

Q1. Wireshark capture: one packet from each layer

Capture traffic while refreshing the Flipkart homepage, then use these filters and header clues to find one packet per layer.

Layer	Wireshark filter / where to look	What to identify in the header
Application	dns or tls / http	DNS query for www.flipkart.com, or a TLS 'Client Hello' / 'Application Data' packet (HTTPS).
Transport	tcp.port == 443	TCP header: Source Port (random, e.g. 52341), Destination Port 443, Sequence No., Flags (SYN, ACK).
Internet	ip.addr == <flipkart IP>	IPv4 header: Source IP (your device), Destination IP (Flipkart server), TTL, Protocol = TCP (6).
Network Access	Open 'Ethernet II' in packet details	Source and Destination MAC addresses, Type = IPv4 (0x0800).

Note: I can't run Wireshark from here, so take screenshots of your own capture for each layer and paste them next to this table. Click the packet, then expand the matching section in the 'Packet Details' pane (Frame, Ethernet II, Internet Protocol, TCP, TLS/HTTP).

Q2. OSI vs TCP/IP model with real-world examples

OSI Layer	TCP/IP Layer	Real-world example (app)
7. Application	Application	Zomato app sends an HTTPS request to show restaurants; WhatsApp uses its messaging protocol.
6. Presentation	Application	Message encrypted (TLS) and photo compressed to JPEG before sending on WhatsApp.
5. Session	Application	You stay logged in to Instagram; the session is kept alive while you scroll.
4. Transport	Transport	TCP splits your Zomato order data into segments, adds port 443, and re-sends lost ones.
3. Network	Internet	IP adds your phone's IP and the Zomato server's IP so routers can forward the packet.
2. Data Link	Network Access	Your Wi-Fi router receives the frame using MAC addresses.
1. Physical	Network Access	Radio signals over Wi-Fi / 4G / 5G carry the bits.

Key point: OSI has 7 layers, TCP/IP has 4. TCP/IP merges the top three OSI layers into Application and the bottom two into Network Access.

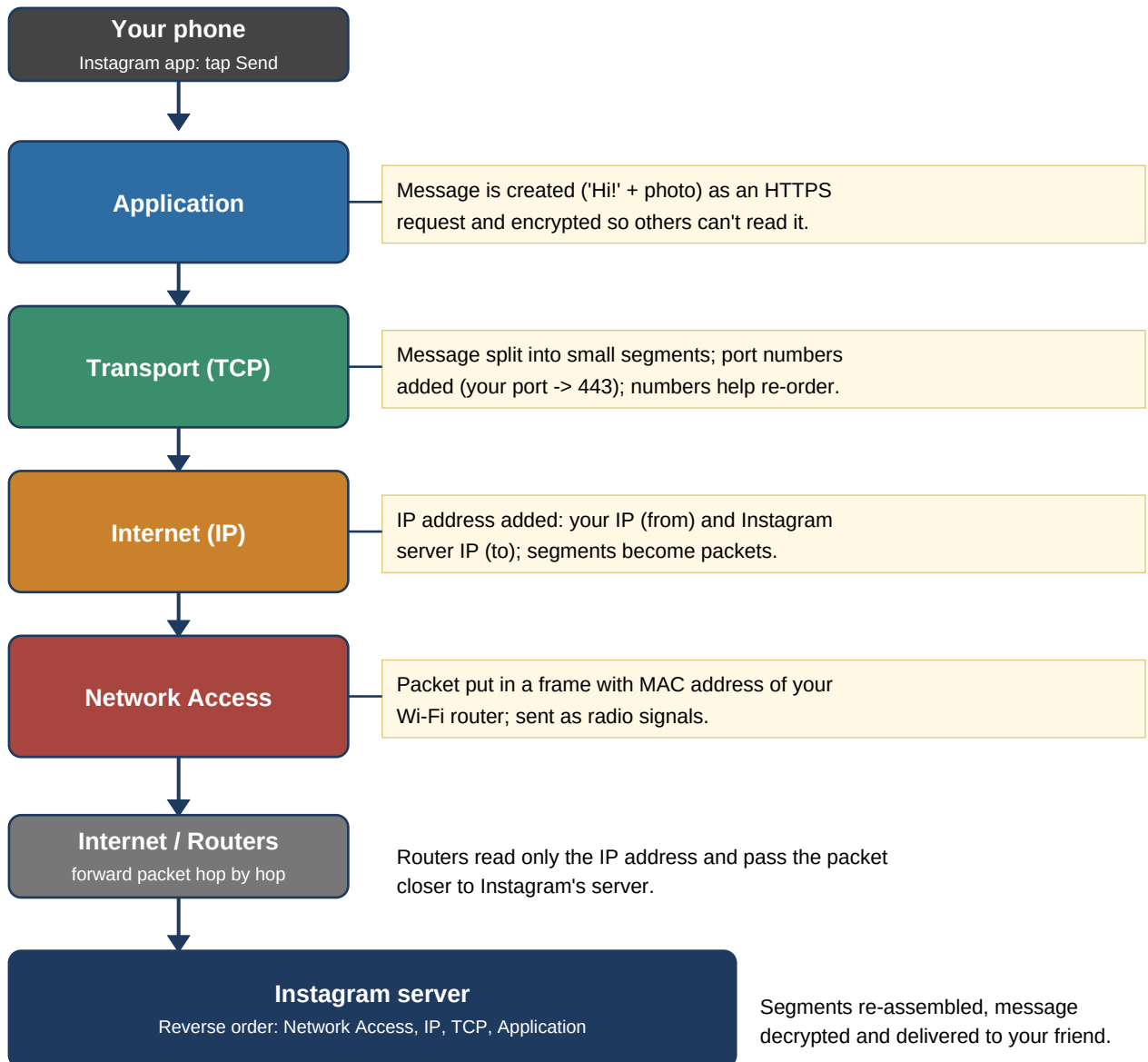
Q3. TCP segment header fields (Transport layer)

The trainer's screenshot was not included in the file, so these are the standard TCP header fields. Match them to your screenshot. All fields in the TCP header belong to the Transport layer.

Field (size)	What it does
Source Port (16 bits)	Port number of the sending app on your device, so replies come back to the right app.
Destination Port (16 bits)	Port of the service on the server (e.g. 443 for HTTPS, 80 for HTTP).
Sequence Number (32 bits)	Numbers the first byte of data in this segment so the receiver can put data in the right order.
Acknowledgment Number (32 bits)	Tells the sender the next byte number the receiver expects, confirming data received.
Data Offset / Header Length (4 bits)	Gives the size of the TCP header, showing where the data begins.
Reserved (3-4 bits)	Kept for future use; set to zero.
Flags (SYN, ACK, FIN, RST, PSH, URG)	Control bits that open (SYN), confirm (ACK), close (FIN) or reset (RST) a connection.
Window Size (16 bits)	How much data the receiver can accept right now; used for flow control.
Checksum (16 bits)	Detects errors in the header and data during transmission.
Urgent Pointer (16 bits)	Marks urgent data when the URG flag is set (rarely used).
Options (variable)	Extra settings such as Maximum Segment Size and window scaling.

Q4. Message from your device to Instagram's server

Diagram: how a message travels down the TCP/IP layers on your phone, across the internet, and up the layers on Instagram's server.



Q5. Analysing one captured packet

Sample answer (replace the values with your own packet's details from Wireshark):

Question	Answer
Packet chosen	A TCP packet from my laptop to the Flipkart server, e.g. 'Frame 125: 74 bytes, [SYN] 52341 -> 443'.
TCP/IP layer	Transport layer (TCP).
How I identified it	Expanded the 'Transmission Control Protocol' section. It shows Source Port 52341, Destination Port 443 (HTTPS), Sequence Number and the SYN flag. Ports, sequence numbers and flags are Transport-layer information. The IP section above it showed the Source and Destination IP addresses (Internet layer).
Real-world action	The SYN packet is the first step of the TCP three-way handshake (SYN, SYN-ACK, ACK). It happens when I refresh the Flipkart homepage: my browser opens a connection to Flipkart before any product data is requested. The same step happens when sending a WhatsApp message or booking a ticket on BookMyShow.

Tip: If you pick a different packet (e.g. DNS = Application layer, or an Ethernet frame = Network Access layer), change the layer and 'how I identified it' accordingly.